MACHINE LEARNING IN PHYSICS
FOUNDATIONS 3

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Recap

What have we learned so far?

1. Machine learning models are trained by minimizing the empirical risk (aka average loss):

$$R(\omega) = \frac{1}{N}\sum_{i=1}^{N} L(y_i, f_i)$$

2. The loss function, $L(y, f)$, is chosen according to the task at hand.

Recap

3. The empirical risk (which is typically referred to as the "loss" in ML circles) is an *unbiased* Monte Carlo estimate of the risk functional

$$R[f] = \int dx \int dy\, L(y, f)\, p(x, y)$$

where $p(x, y)$ is the probability density of the data.

4. Minimizing the risk functional with respect to the ML model f shows that the optimal model satisfies

$$\int dy \frac{\partial L}{\partial f} p(y \mid x) = 0$$

EMPIRICAL RISK MINIMIZATION

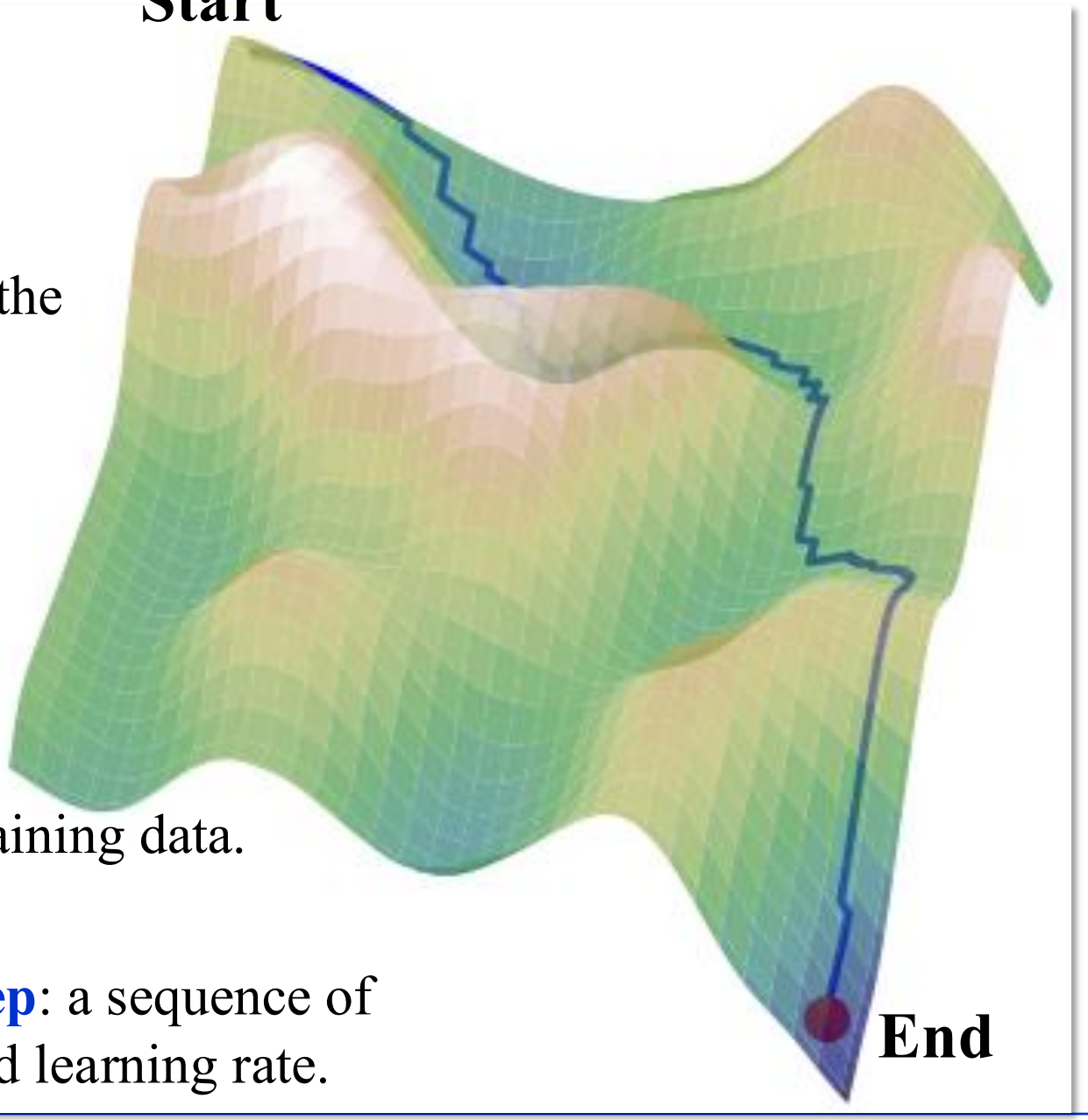
Some jargon

Batch: a subset of the training data.

Iteration: a single optimization step.

Epoch: a single pass through the training data.

Learning Rate Step: a sequence of iterations with fixed learning rate.



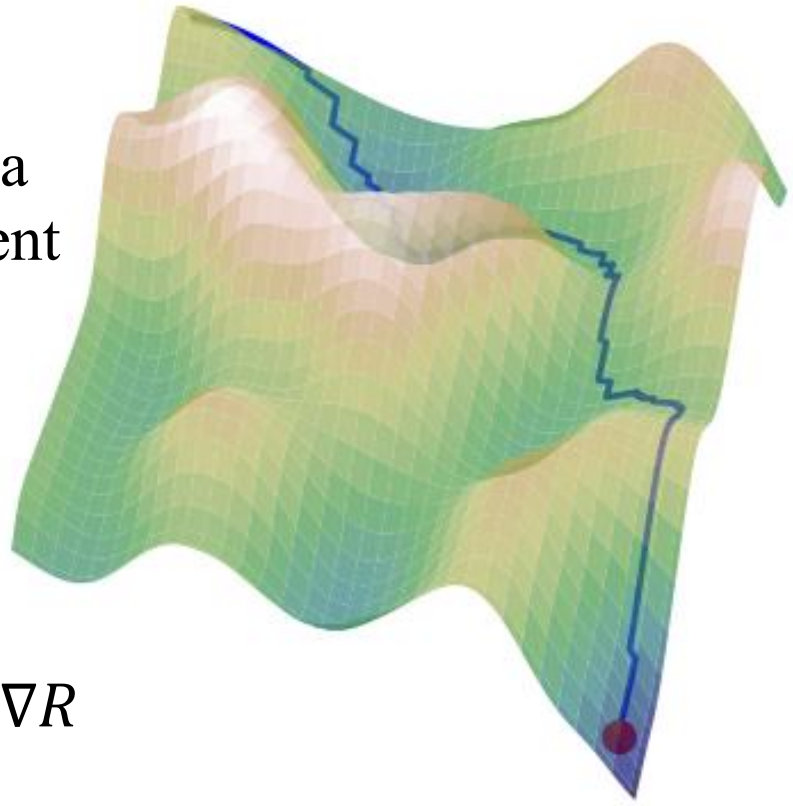
Empirical Risk Minimization (1)

The minimization of $R(\omega)$ is typically done by moving in the direction of *steepest descent*. Most optimization algorithms in ML are variations of **Stochastic Gradient Descent**.

Algorithm

1. At the current point ω_j, compute a *noisy* approximation of the gradient of $R(\omega) \approx \frac{1}{n}\sum_{i=1}^{n} L(y_i, f_i)$ by using a **batch** of **training data**, where $\boldsymbol{n \ll N}$.
2. Move to the next position ω_{j+1} in the landscape using

$$\omega_{j+1} = \omega_j - \eta \nabla R$$



Empirical Risk Minimization (2)

Why does this (1st-order) algorithm

$$\omega_{j+1} = \omega_j - \eta \nabla R$$

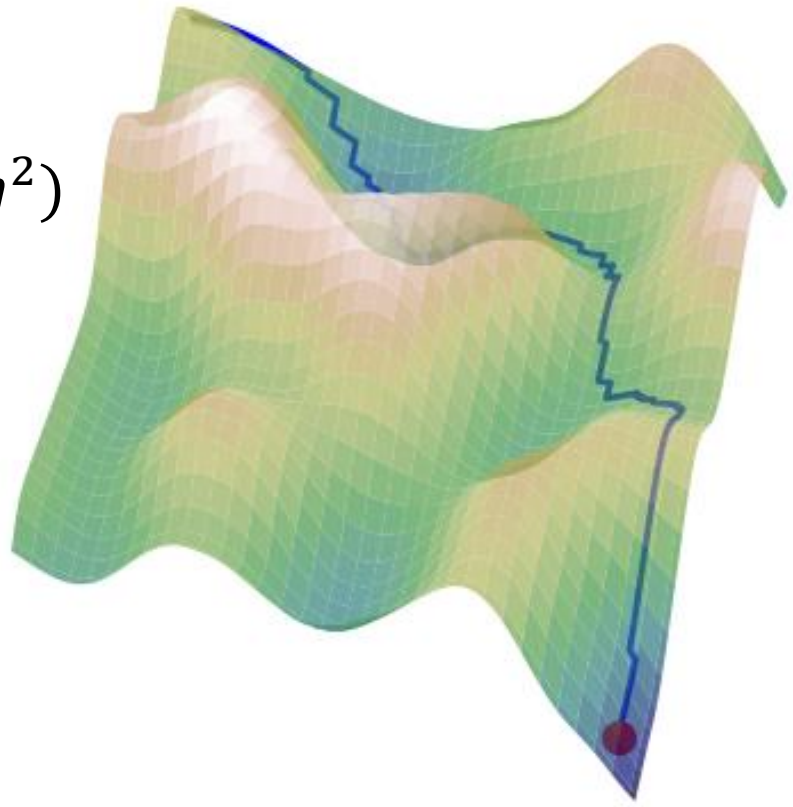
work?

Consider

$$R(\omega_{j+1}) = R(\omega_j - \eta \nabla R)$$
$$= R(\omega_j) - \eta \nabla R \cdot \nabla R + O(\eta^2)$$

If the $O(\eta^2)$ can be neglected, and given that the $O(\eta)$ term is always negative, it follows that

$$R(\omega_{j+1}) < R(\omega_j).$$



BINARY CLASSIFIERS

Binary Classifiers (1)

Starting with

$$\int dy \frac{\partial L}{\partial f}\, p(y|x) = 0$$

it was shown in Lecture 2 and tutorial 2 that a binary classifier $f(x, \omega)$ trained using the cross-entropy loss $L(y, f) = -[y \log f + (1 - y) \log(1 - f)]$ approximates the **class probability**

$$f(x, \omega^*) = p(1|x)$$

that is, the probability that an object characterized by the **features** x belongs to the class labeled $y = 1$.

Binary Classifiers (2)

From Bayes' theorem, we can write the class probability as

$$p(1|x) = \frac{p(x|1)\,\pi(1)}{p(x)}$$

where $p(x) = p(x|1)\,\pi(1) + p(x|0)\,\pi(0)$ and

- $p(x|k)$ is the probability density of x for class $y = k$,
- $\pi(k)$ is the **prior probability** of class $y = k$, and
- $\pi(1) + \pi(0) = 1.$

Binary Classifiers (3)

When the training dataset is **balanced** (as in tutorial 2), that is, $\pi(1) = \pi(0)$, the class probability $p(1|x)$ becomes the **discriminant**,

$$D(x) = \frac{p(x|1)}{p(x|1) + p(x|0)}$$

This is one of the most useful results in scientific machine learning.

Binary Classifiers (4)

It is possible to train a binary classifier using an **unbalanced** dataset: $\pi(1) \neq \pi(0)$.

But if $\pi(1) \ll \pi(0)$ this might be challenging.

Why do you think this is?

However, as shown on the next slide, $p(1|x)$ can be computed from a model trained with a balanced dataset.

Binary Classifiers (4)

According to Bayes' theorem

$$p(1|x) = \frac{p(x|1)\pi(1)}{p(x|1)\pi(1) + p(x|0)\pi(0)}$$

Dividing the numerator and denominator by $p(x|1) + p(x|0)$ yields

$$p(1|x) = \frac{D(x)\pi(1)}{D(x)\pi(1) + [1 - D(x)]\pi(0)}$$

So, it is not necessary to use the true priors during training.

CLASSIFIER PERFORMANCE

Classifier Performance (1)

Consider a classifier, $D(x)$, the indicator function $I[*]$, the targets y, and a threshold t on $D(x)$ above which one is prepared to assign x to the "positive" (signal) class and below which one assigns it to the "negative" (background) class.

The following counts can be computed:

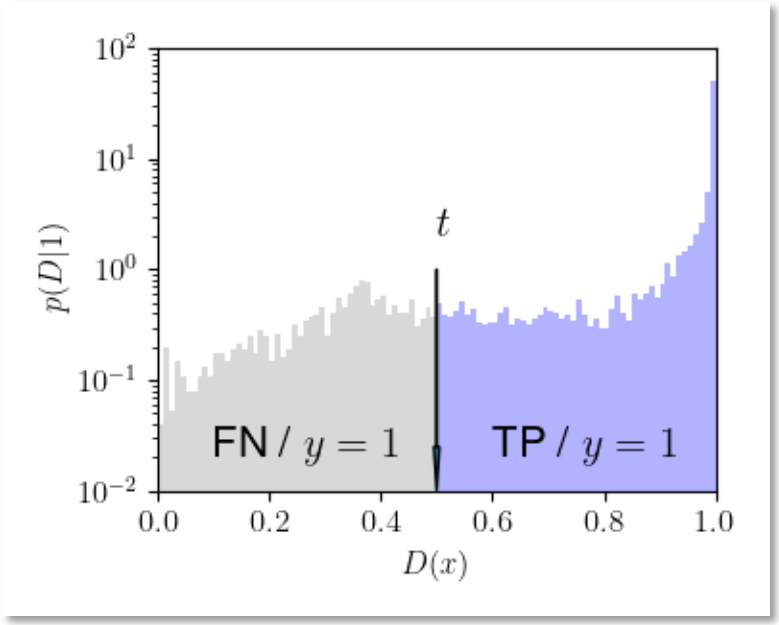
1. False Negatives (FN): $\sum_{i,\, y_i=1} I[D(x_i) \leq t]$
2. True Positives (TP): $\sum_{i,\, y_i=1} I[D(x_i) > t]$
3. True Negatives (TN): $\sum_{i,\, y_i=0} I[D(x_i) \leq t]$
4. False Positives (FP): $\sum_{i,\, y_i=0} I[D(x_i) > t]$

$I[z] = 1$ if z is true, 0 otherwise.

Classifier Performance (2)

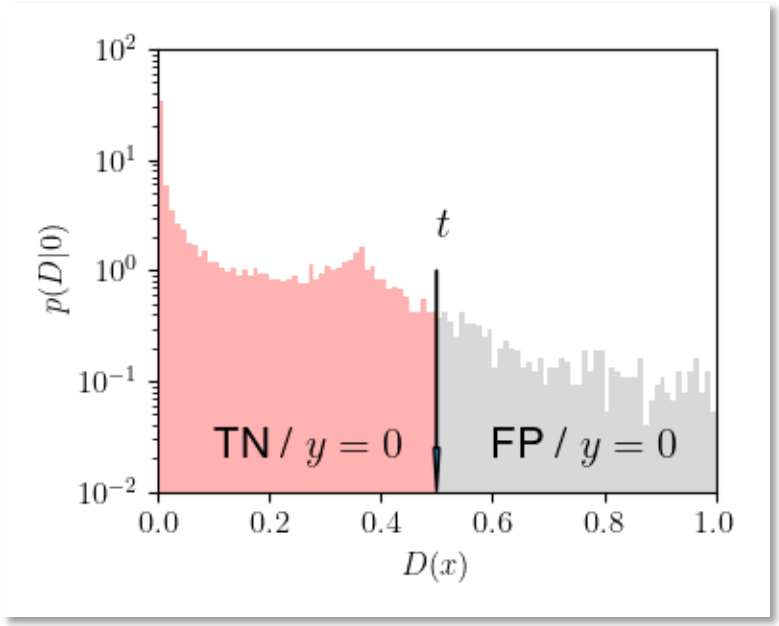
FN: $\sum_{i,\, y_i=1} I[D(x_i) \leq t]$ $t = 0.5$

TP: $\sum_{i,\, y_i=1} I[D(x_i) > t]$



$y = 1$

$y = 0$



TN: $\sum_{i,\, y_i=0} I[D(x_i) \leq t]$

FP: $\sum_{i,\, y_i=0} I[D(x_i) > t]$

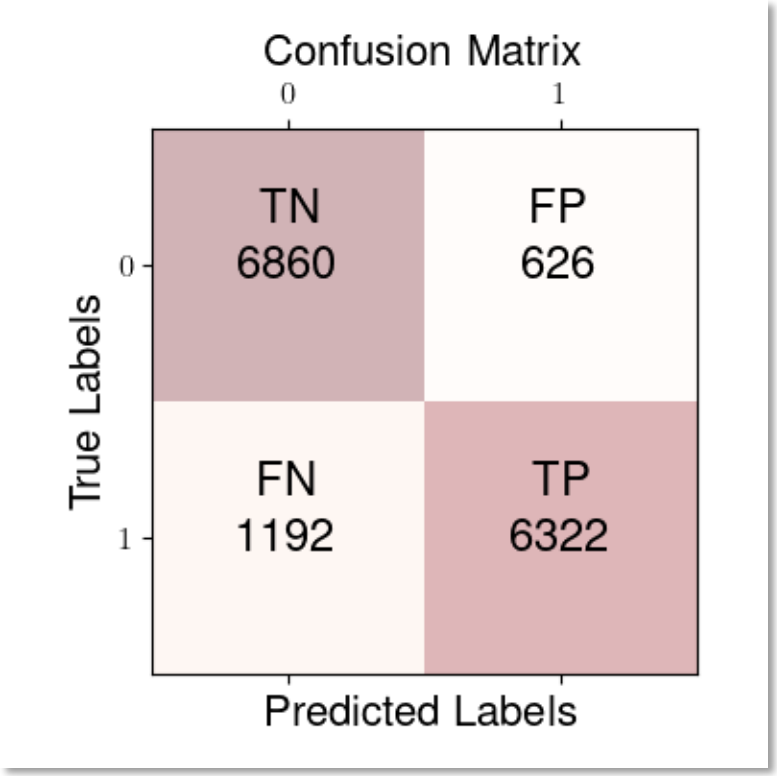
Confusion Matrix

FN: $\sum_{i,\, y_i=1} I[D(x_i) \leq t]$

TP: $\sum_{i,\, y_i=1} I[D(x_i) > t]$

The confusion matrix is a simple way to summarize the counts in the four disjoint regions.

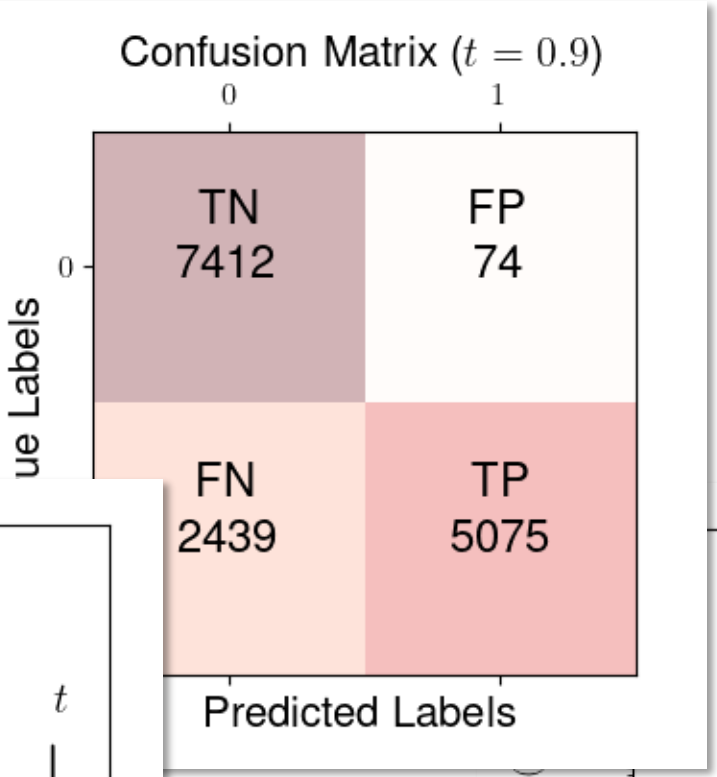
A perfectly separable dataset would yield a diagonal matrix.

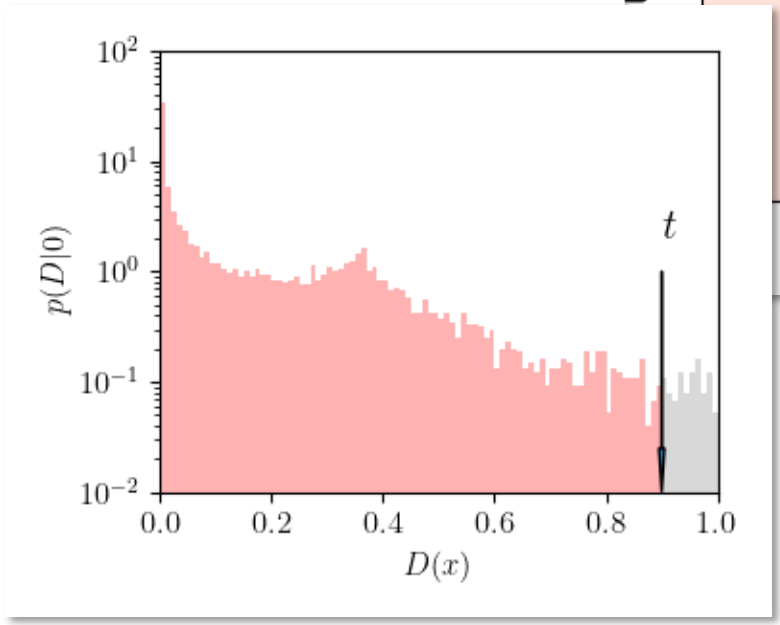


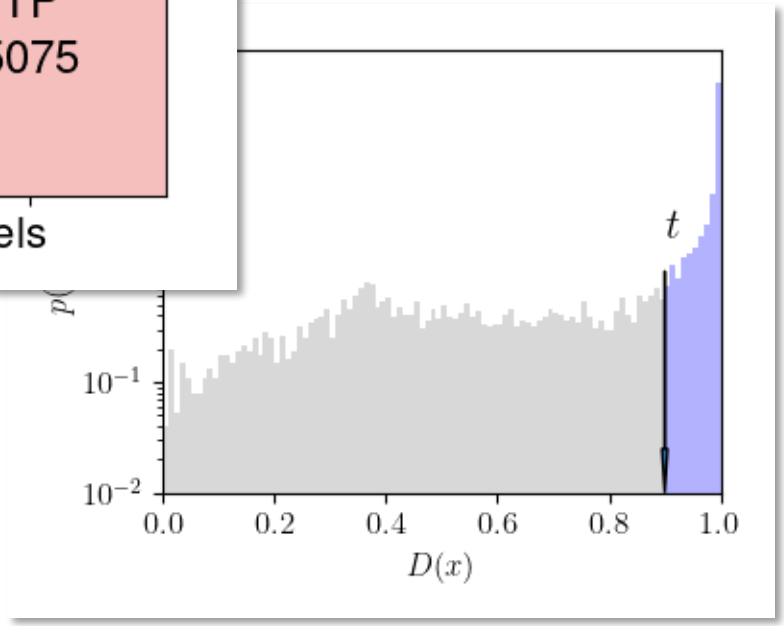
TN: $\sum_{i,\, y_i=0} I[D(x_i) \leq t]$

FP: $\sum_{i,\, y_i=0} I[D(x_i) > t]$

Confusion Matrix ($t = 0.9$)







Accuracy, Precision, Recall

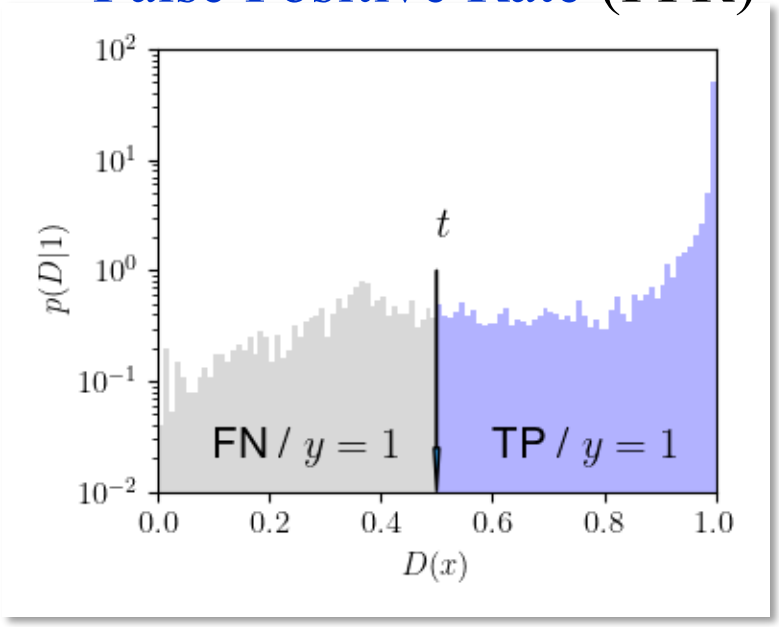
Accuracy = (TP + TN) / (TP + TN + FN + FP)

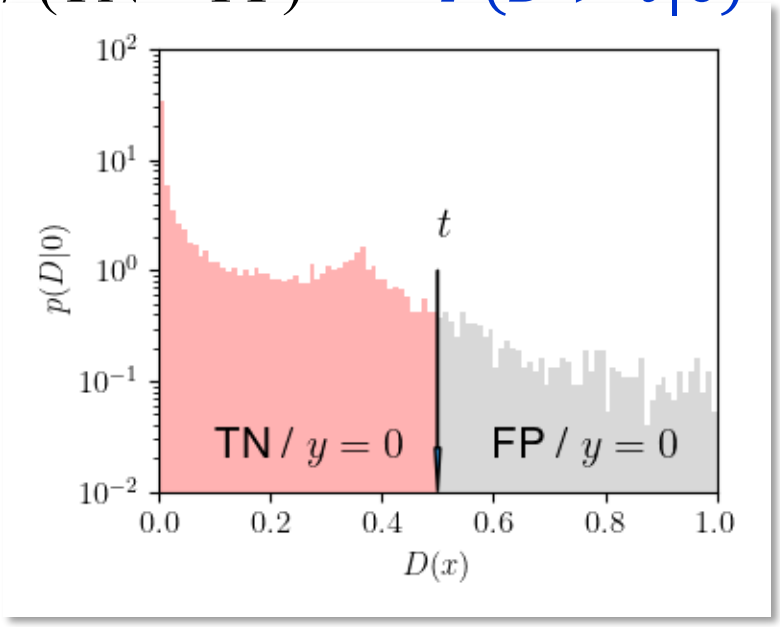
Precision = TP / (TP + FP) $P(y = 1|D > t)$

Recall = TP / (FN + TP)

True Positive Rate (TPR) = TP / (FN + TP) $P(D > t|1)$

False Positive Rate (FPR) = FP / (TN + FP) $P(D > t|0)$





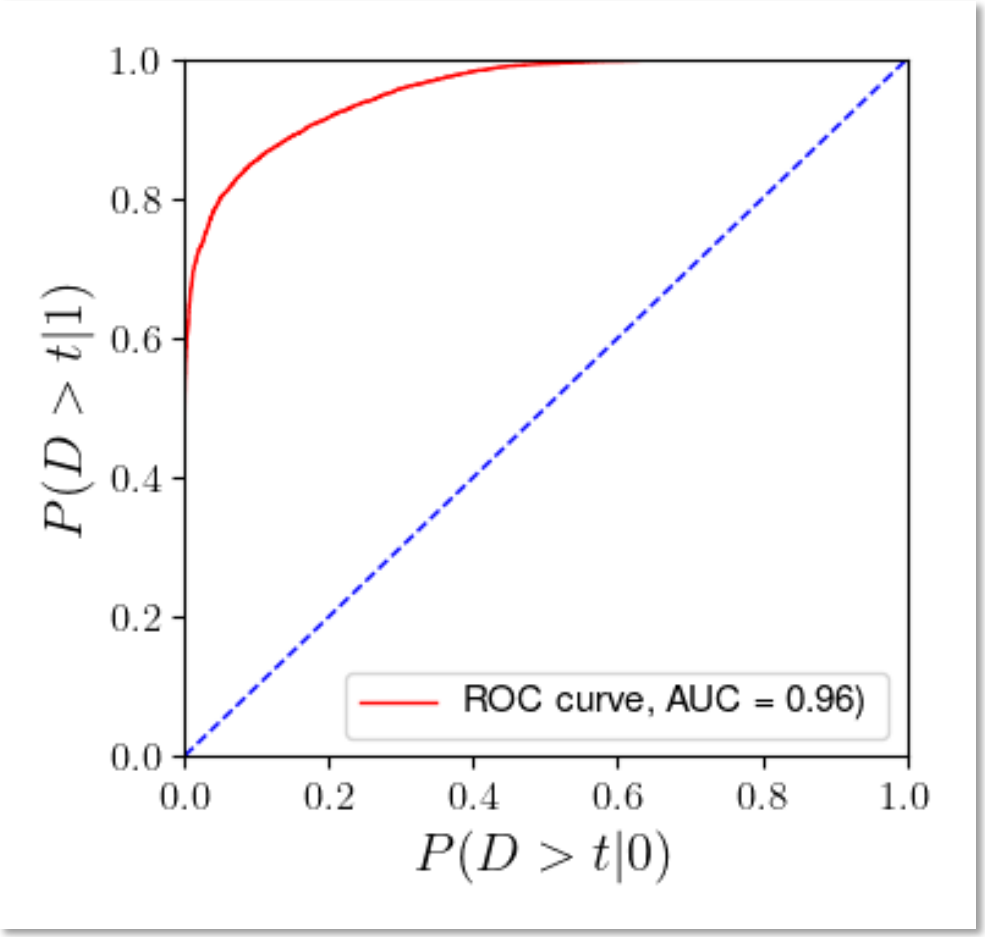
ROC and AUC

The ROC curve is a plot of TPR vs. FPR, that is,

$P(D > t|1)$ vs. $P(D > t|0)$

where (usually)

$$D(x) = \frac{p(x|1)}{p(x|1) + p(x|0)}$$



BEYOND CLASSIFICATION

Beyond Classification (1)

Notice that the discriminant

$$D(x) = \frac{p(x|1)}{p(x|1) + p(x|0)}$$

can be rewritten as

$$p(x|1) = \left[\frac{D(x)}{1 - D(x)}\right] p(x|0)$$

The above is often referred to as the likelihood ratio trick.

If $p(x|0)$ is a *known* density from which the $y = 0$ vectors are sampled, then the discriminant can be repurposed as a **probability density estimator**.

Beyond Classification (2)

Consider a dataset $\mathrm{U}[x]$, of size N, labeled $y = 1$. The goal is to approximate its associated probability density $u(x)$.

Algorithm

1. Generate another dataset $\mathrm{K}[x]$, of size N, labeled $y = 0$, where $x \sim k(x) \equiv p(x|0)$, a *known* density*.

2. Train a classifier $D(x)$ using a randomized mixture of the datasets $\mathrm{U}[x]$ and $\mathrm{K}[x]$ of sample size $2N$.
3. Approximate the unknown density $u(x)$ using

$$u(x) = k(x)\left[\frac{D(x)}{1 - D(x)}\right]$$

* The symbol "~" means "sampled from".

Beyond Classification (3)

But we can go further*.

Suppose our dataset U of sample size N is from a simulation that depends on a set of parameters θ.

Let's sample these parameters from a *known* prior, $\pi(\theta)$. For each sampled point θ, x is sampled from the *unknown* density $u(x|\theta)$ using the simulator.

The dataset, labeled $y = 1$, comprises N pairs $\{(x, \theta)\}$ that constitute a **point cloud** representation of the joint density $u(x, \theta)$.

* Baldi, P., Cranmer, K., Faucett, T. *et al.* Parameterized neural networks for high-energy physics. *Eur. Phys. J. C* **76**, 235 (2016).

Beyond Classification (4)

Algorithm

1. Generate a dataset K of size N, labeled $y = 0$, where θ is sampled from the *same* prior $\pi(\theta)$. For each θ, sample $x \sim k(x|\theta)$, where $k(x|\theta)$ is a *known* conditional density.

2. Train a classifier $f(x, \theta, \omega)$ with an admixture of U and K.

3. Compute $u(x, \theta) = k(x|\theta)\pi(\theta)\left[\frac{D(x,\theta)}{1-D(x,\theta)}\right]$.

4. Divide by $\pi(\theta)$:

$$u(x|\theta) = k(x|\theta)\left[\frac{D(x,\theta)}{1-D(x,\theta)}\right]$$

Beyond Classification (5)

The ratio

$$\frac{D(x,\theta)}{1-D(x,\theta)}$$

could have a large dynamic range.

If so, instead of approximating $D(x,\theta)$, it may be better to use the appropriate* loss function to approximate

$$f(x,\theta) = \log\left[\frac{D(x,\theta)}{1-D(x,\theta)}\right]$$

and then compute

$$\log u(x,\theta) = \log k(x|\theta) + \log \pi(\theta) + f(x,\theta)$$
$$\log u(x|\theta) = \log u(x,\theta) - \log \pi(\theta)$$

* Which one?

Summary

- In addition to the ROC curve and the AUC, the performance of a classifier can be characterized with several other numbers including:
  - False Negatives (FN)
  - True Positives (TP)
  - True Negatives (TN)
  - False Positives (FP)

- A classifier can be repurposed in many ways, including as a probability density estimator.